

Tribhuvan University  
Faculty of Humanities & Social Sciences  
OFFICE OF THE DEAN

2023

## Bachelor in Computer Applications

Course Title: **Network Programming**

Code No: **CACS355** Semester: **VI**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

*Candidates are required to answer the questions in their own words as far as practicable.*

**Note:** Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

### Group B

2. Define the client-server model of a network application. Explain the design considerations for a network application.
3. Write a program to show the IP and MAC address of the system and also check if the IP address is IPv4 or IPv6.
4. Define URL Connection Class? Write a program to read header using header-specific methods.
5. Write the functions of Socket. Explain the configuration options of a TCP client socket.
6. Define a secure socket. Write a program to create a secure client socket.
7. Define Buffer. Explain buffer handling methods.
8. Write short notes on (any two): a) HTTP methods b) Datagram Packet c) Multicast Socket

### Group C

*Attempt any TWO questions [2x10=20]*

9. a) Differentiate between URL and URI class with their applications. b) Write any program to illustrate the application of RMI.
10. Write a TCP multithreaded server and client socket program to check the number sent by the client is either prime or composite.
11. Write a program to create a UDP server and client which converts the date from the client in BS to AD format. [Assume that the difference between BS and AD is 56 years, 08 months, 17 days]